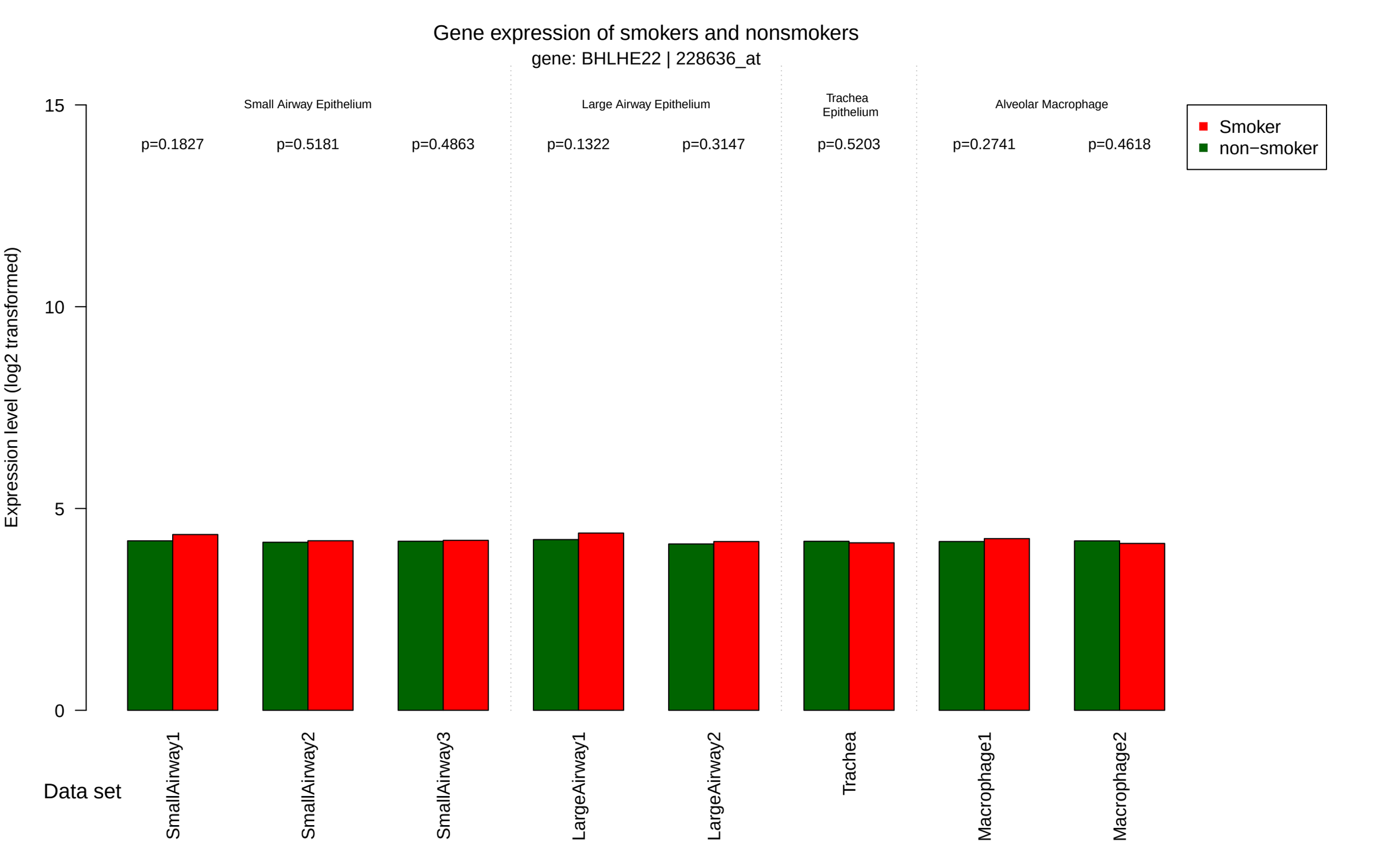
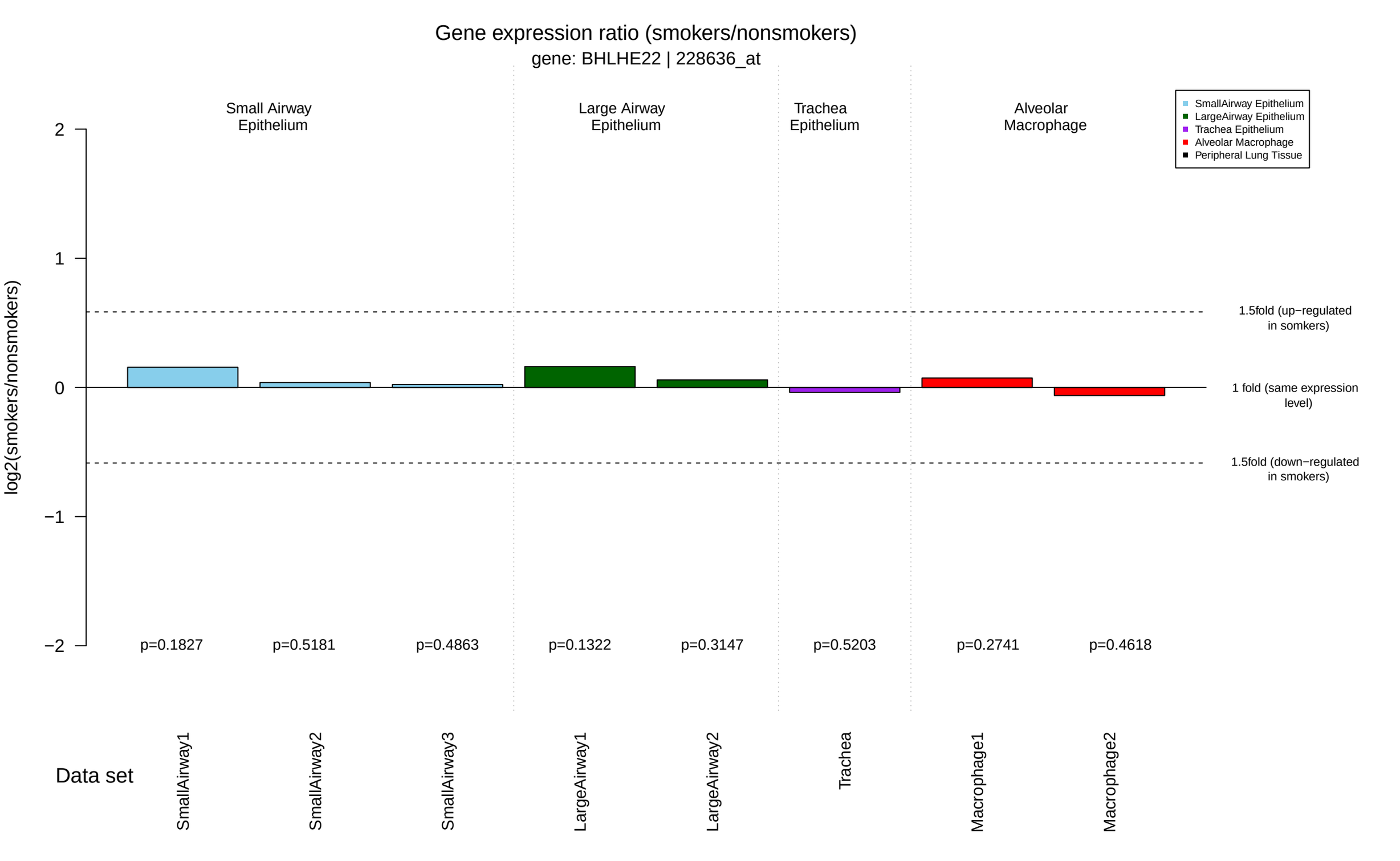
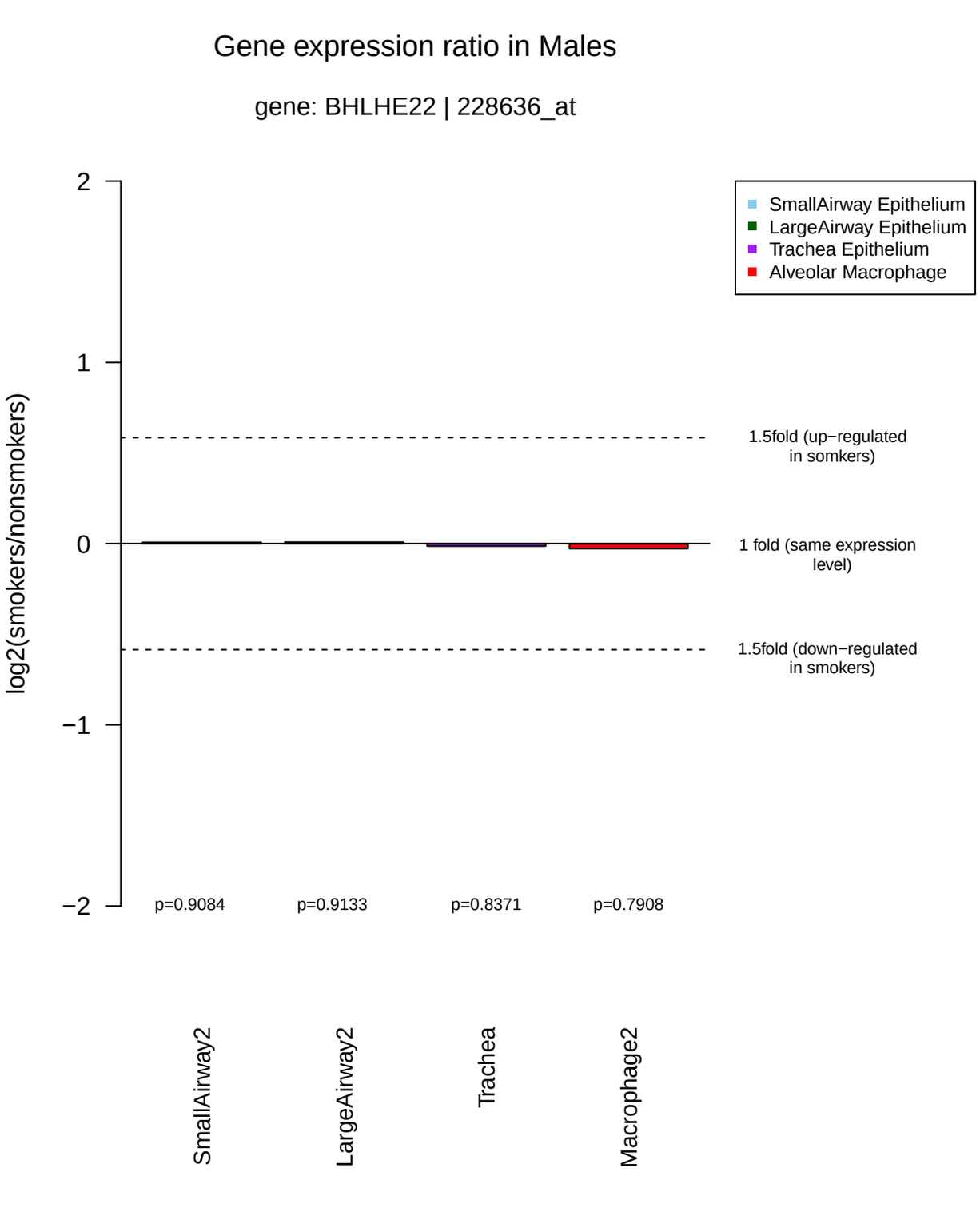
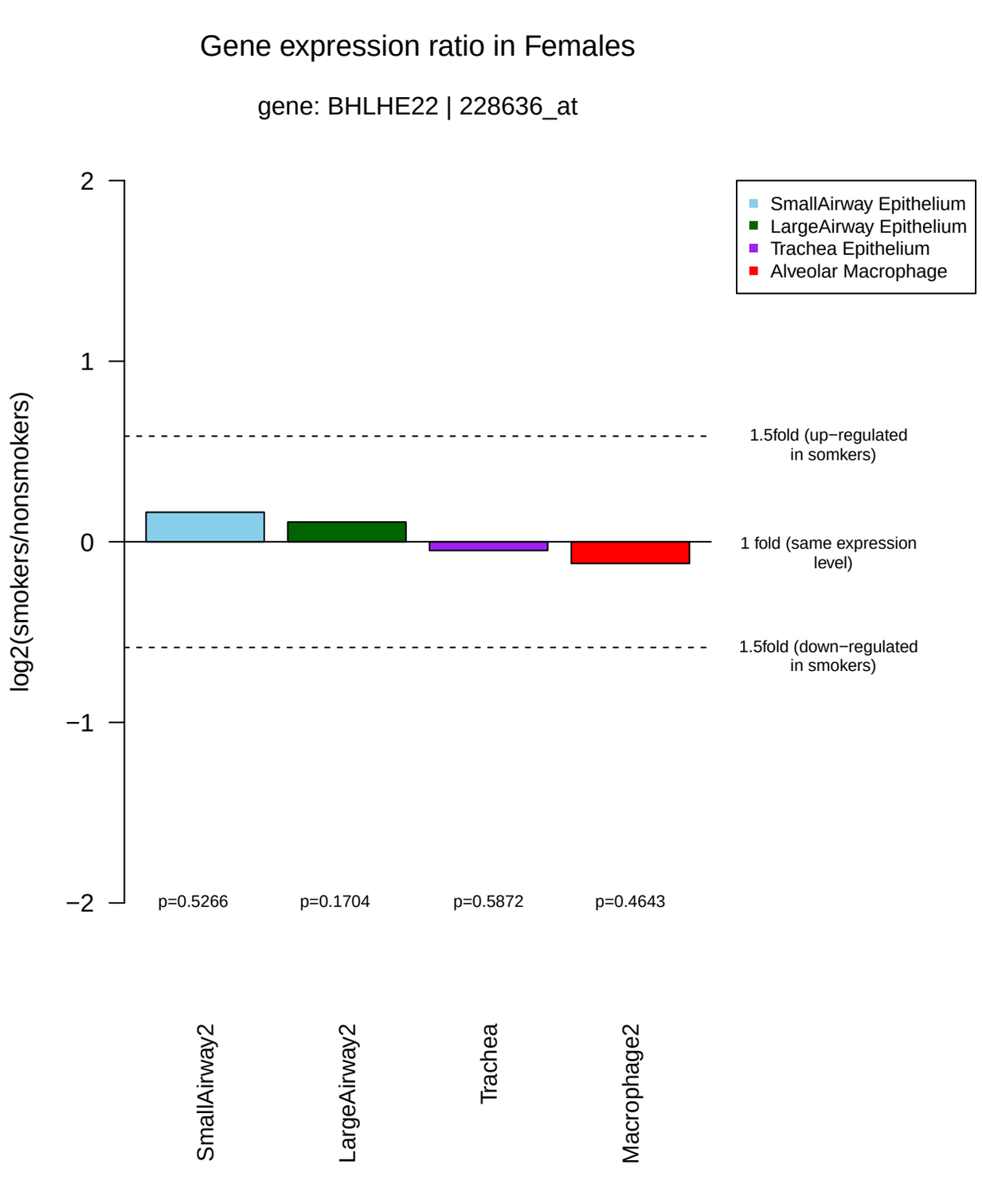
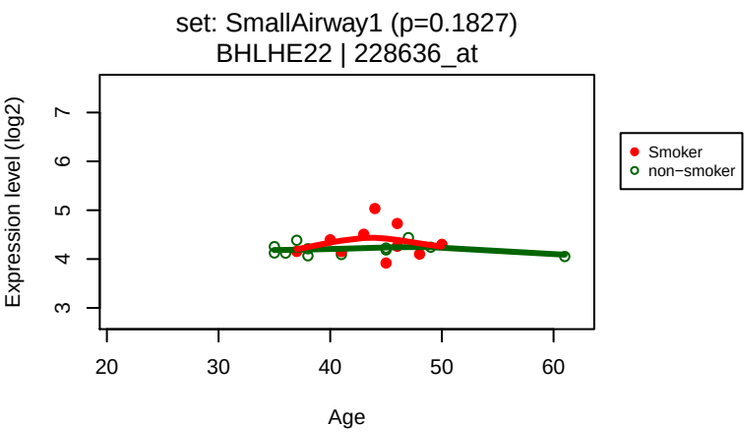
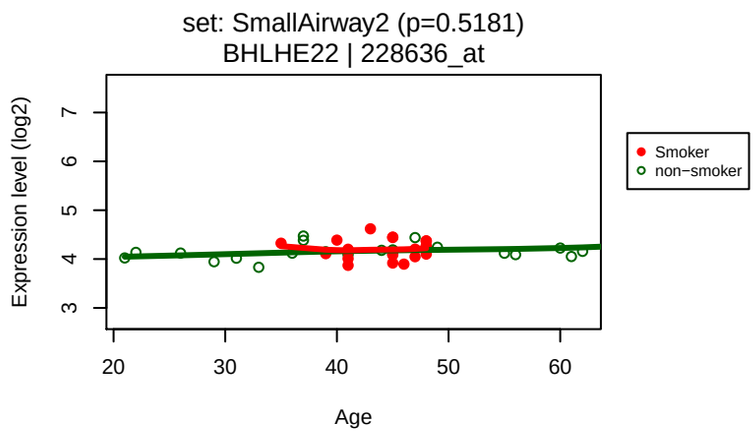

Gene expression of smokers and nonsmokers
gene: BHLHE22 | 228636_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1827
p=0.5181
p=0.4863
p=0.1322
p=0.3147
p=0.5203
p=0.2741
p=0.4618
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

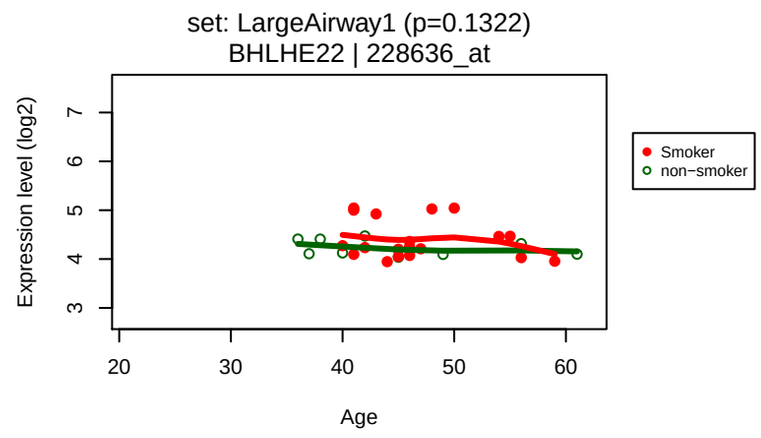

Gene expression ratio (smokers/nonsmokers)
gene: BHLHE22 | 228636_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1827
p=0.5181
p=0.4863
p=0.1322
p=0.3147
p=0.5203
p=0.2741
p=0.4618
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

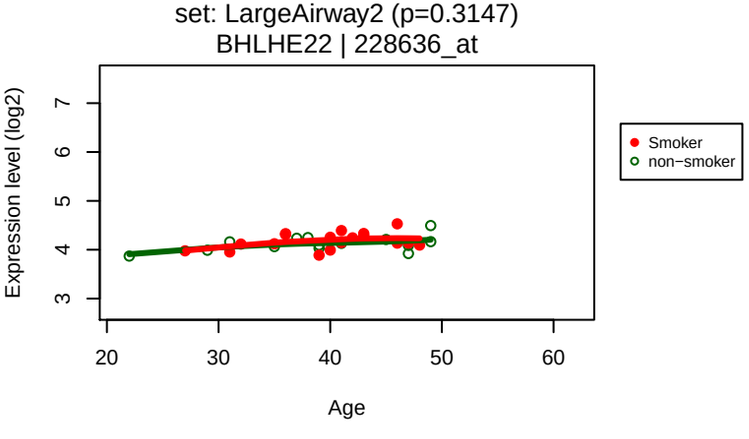

Gene expression ratio in Males
gene: BHLHE22 | 228636_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9084
p=0.9133
p=0.8371
p=0.7908
SmallAirway2
LargeAirway2
Trachea
Macrophage2

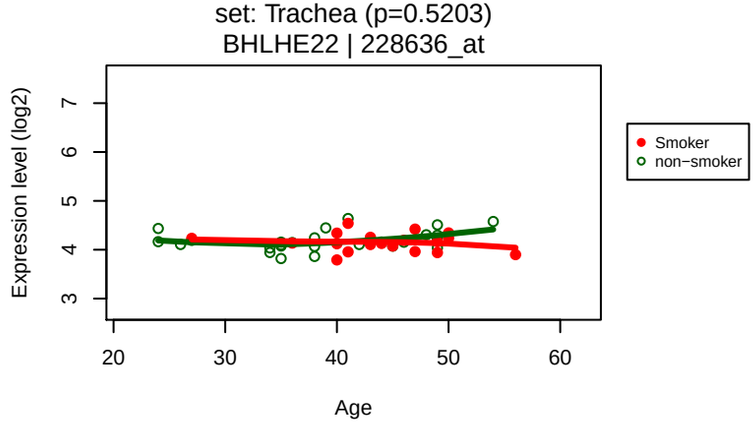

Gene expression ratio in Females
gene: BHLHE22 | 228636_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5266
p=0.1704
p=0.5872
p=0.4643
SmallAirway2
LargeAirway2
Trachea
Macrophage2

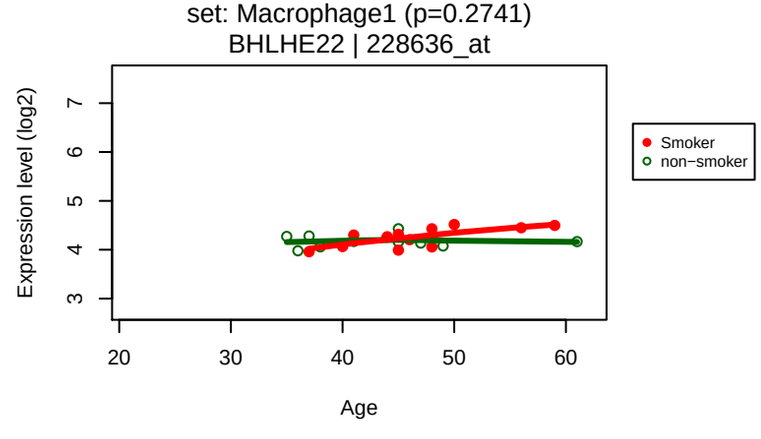

set: SmallAirway1 (p=0.1827)
BHLHE22 | 228636_at
Expression level (log2)
Age
Smoker
non−smoker

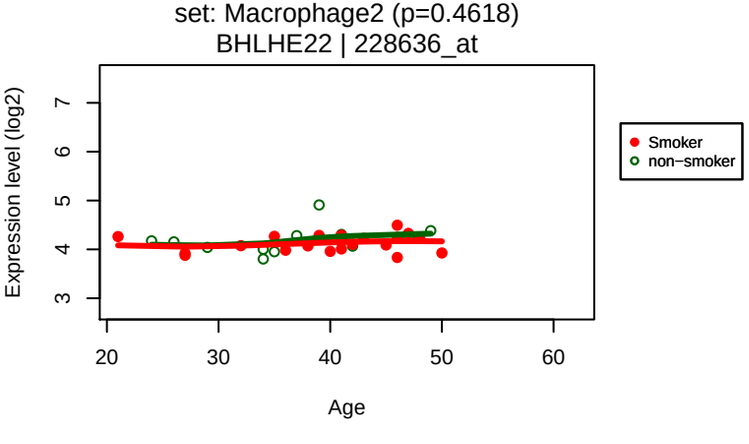

set: SmallAirway2 (p=0.5181)
BHLHE22 | 228636_at
Expression level (log2)
Age
Smoker
non−smoker

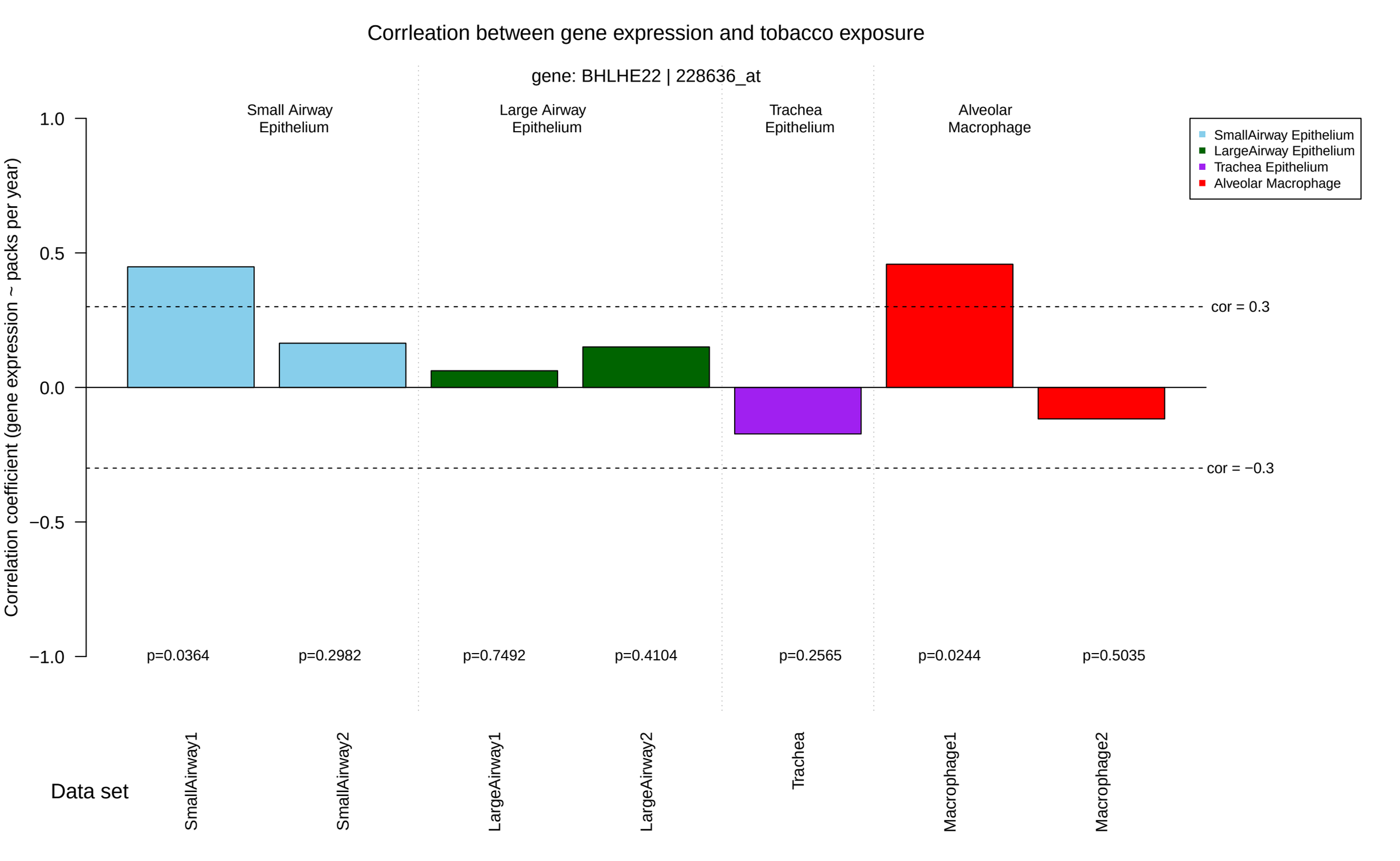

set: LargeAirway1 (p=0.1322)
BHLHE22 | 228636_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3147)
BHLHE22 | 228636_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5203)
BHLHE22 | 228636_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2741)
BHLHE22 | 228636_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4618)
BHLHE22 | 228636_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BHLHE22 | 228636_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0364
p=0.2982
p=0.7492
p=0.4104
p=0.2565
p=0.0244
p=0.5035
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2